

Automata Theory MCQs (1–50)

1. A Turing Machine is formally represented as:
A) $(Q, \Sigma, \delta, q_0, F)$
B) $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$
C) (Q, Σ, P, S)
D) $(Q, \Sigma, \Delta, \lambda)$
2. Which of the following has the highest computational power?
A) DFA
B) NFA
C) PDA
D) TM
3. A Universal Turing Machine can:
A) Accept only regular languages
B) Simulate any Turing Machine
C) Accept only CFLs
D) Minimize DFAs
4. Which memory structure is used by a PDA?
A) Queue
B) Stack
C) Tape
D) Array
5. The language accepted by DFA belongs to:
A) CFL
B) CSL
C) Regular Language
D) RE
6. The transition function of a DFA is:
A) $\delta: Q \times \Sigma \rightarrow 2^Q$
B) $\delta: Q \times \Sigma \rightarrow Q$
C) $\delta: Q \times \Gamma \rightarrow Q$
D) $\delta: Q \rightarrow \Sigma$
7. In an NFA, for a state and input symbol there can be:
A) Exactly one transition
B) At most one transition
C) Multiple transitions
D) No transitions only
8. Which move is allowed in ϵ -NFA?
A) λ -move
B) Stack move
C) Tape move
D) Left move
9. Every NFA has:
A) No equivalent DFA
B) Exactly one DFA

- C) An equivalent DFA
 - D) A PDA
10. Which of the following is NOT a component of CFG?
- A) Variables
 - B) Terminals
 - C) Productions
 - D) Tape
11. A CFG is represented as:
- A) $(Q, \Sigma, \delta, q_0, F)$
 - B) (V, T, P, S)
 - C) (Q, Γ, δ)
 - D) (Σ, T, P)
12. The grammar $S \rightarrow aSb \mid \epsilon$ generates:
- A) a^n
 - B) b^n
 - C) $a^n b^n$
 - D) $(ab)^n$
13. A grammar whose productions are either left-linear or right-linear is:
- A) CFG
 - B) CSG
 - C) Regular Grammar
 - D) Unrestricted Grammar
14. Which grammar corresponds to Type-3 in Chomsky hierarchy?
- A) CFG
 - B) CSG
 - C) Regular Grammar
 - D) Unrestricted Grammar
15. The root node of a parse tree is:
- A) Terminal
 - B) Variable
 - C) Start Symbol
 - D) Stack Symbol
16. An ambiguous grammar generates:
- A) No string
 - B) Exactly one parse tree
 - C) More than one parse tree for a string
 - D) Only regular languages
17. The language of all strings over Σ including ϵ is:
- A) Σ^+
 - B) Σ^*
 - C) $\emptyset$
 - D) $\{\epsilon\}$
18. Σ^+ equals:
- A) $\Sigma^* \cup \{\epsilon\}$
 - B) $\Sigma^* - \{\epsilon\}$
 - C) Σ^*
 - D) $\emptyset$

19. Kleene Star allows:
- A) One repetition
 - B) One or more repetitions
 - C) Zero or more repetitions
 - D) Two or more repetitions
20. Kleene Plus allows:
- A) Zero repetitions
 - B) One or more repetitions
 - C) Exactly one repetition
 - D) Infinite repetitions only
21. Which relation is correct?
- A) $L^+ = L^*$
 - B) $L^* = L^+$
 - C) $L^* = L^+ \cup \{\epsilon\}$
 - D) $L^+ = L^* \cup \{\epsilon\}$
22. The length of ϵ is:
- A) 1
 - B) 2
 - C) -1
 - D) 0
23. The empty set contains:
- A) One string
 - B) ϵ
 - C) No elements
 - D) One symbol
24. Which is true?
- A) $\epsilon = \emptyset$
 - B) $\epsilon \in \emptyset$
 - C) $\epsilon \neq \emptyset$
 - D) $\emptyset$ contains ϵ
25. Regular languages are closed under:
- A) Union only
 - B) Concatenation only
 - C) Kleene Star only
 - D) Union, Concatenation and Kleene Star
26. If $L_1 = \{a\}$ and $L_2 = \{b\}$, then $L_1 L_2 =$
- A) $\{a, b\}$
 - B) $\{ab\}$
 - C) $\{ba\}$
 - D) $\emptyset$
27. A Moore machine generates output based on:
- A) Input only
 - B) State only
 - C) State and Input
 - D) Stack
28. A Mealy machine generates output based on:
- A) State only
 - B) Input only

- C) State and Input
 - D) Tape
29. Which machine generally requires fewer states?
- A) Moore
 - B) Mealy
 - C) PDA
 - D) DFA
30. Output in a Mealy machine is associated with:
- A) States
 - B) Productions
 - C) Transitions
 - D) Stack
31. Output in a Moore machine is associated with:
- A) States
 - B) Inputs
 - C) Productions
 - D) Tape cells
32. The language $\{ww^R\}$ can be accepted by:
- A) DPDA
 - B) DFA
 - C) NPDA
 - D) Moore Machine
33. DPDA accepts:
- A) CFL
 - B) DCFL
 - C) CSL
 - D) RE
34. NPDA accepts:
- A) DCFL
 - B) CFL
 - C) Regular Languages only
 - D) CSL
35. Which statement is correct?
- A) DPDA = NPDA
 - B) DPDA > NPDA
 - C) NPDA > DPDA
 - D) Both are incomparable
36. The Pumping Lemma is mainly used to prove:
- A) A language is regular
 - B) A language is not regular
 - C) A grammar is CFG
 - D) A TM halts
37. The language $\{a^n b^n \mid n \geq 0\}$ is:
- A) Regular
 - B) CFL
 - C) Finite
 - D) Recursive only

38. The language $\{a^n b^n \mid n \geq 0\}$ is not regular because:
- A) Closure property
 - B) Rice's theorem
 - C) Pumping Lemma
 - D) Arden's theorem
39. Which theorem is used to derive a Regular Expression from an FA?
- A) Rice's Theorem
 - B) Arden's Theorem
 - C) Pumping Lemma
 - D) Myhill-Nerode
40. Arden's theorem solves equations of the form:
- A) $R = Q + RP$
 - B) $a^n b^n$
 - C) ww^R
 - D) $CFG \rightarrow PDA$
41. Which problem is undecidable?
- A) DFA Acceptance
 - B) DFA Emptiness
 - C) Halting Problem
 - D) NFA Acceptance
42. $HALT = \{\langle M, w \rangle \mid M \text{ halts on } w\}$ is:
- A) Regular
 - B) Decidable
 - C) Undecidable
 - D) Context-Free
43. Rice's Theorem applies to:
- A) All properties of strings
 - B) Non-trivial properties of RE languages
 - C) Only regular languages
 - D) Only CFLs
44. Which language class is accepted by Turing Machines?
- A) Regular
 - B) CFL
 - C) RE
 - D) DCFL
45. Linear Bounded Automata (LBA) accept:
- A) Regular Languages
 - B) CFLs
 - C) CSLs
 - D) RE Languages
46. The Chomsky hierarchy from lowest to highest power is:
- A) $RG < CFG < CSG < UG$
 - B) $UG < CSG < CFG < RG$
 - C) $CFG < RG < CSG < UG$
 - D) $RG < CSG < CFG < UG$
47. Which automaton corresponds to CFL?
- A) DFA
 - B) NFA

C) PDA

D) TM

48. Which automaton corresponds to CSL?

A) LBA

B) PDA

C) DFA

D) NFA

49. To check equivalence of two automata, we generally:

A) Convert to PDA

B) Convert to CFG

C) Minimize equivalent DFAs and compare

D) Use Pumping Lemma

50. Which statement is correct?

A) $\text{DFA} \supset \text{PDA} \supset \text{TM}$

B) $\text{TM} \supset \text{PDA} \supset \text{DFA}$

C) $\text{PDA} \supset \text{TM} \supset \text{DFA}$

D) $\text{DFA} \supset \text{TM} \supset \text{PDA}$